

SYED ALI

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WORK EXPERIENCE

Machine Learning Researcher, Rusk Media

Feb 2026 – Present

Keywords: Gemini, Multimodal LLMs, Recommender Systems, k-NN Retrieval, Vector Search, Python

Bangalore, India

Building production ML systems for video post-production automation and content recommendation

- Shipped two post-production tools built on multimodal Gemini models: an *SRT Generator* (per-episode subtitle turnaround *2 days* → *25 seconds*) and an *SRT Translator* (full-season translation *100+ hours* → *5 minutes*), each designed end-to-end with a visual, self-serve UI and now handling live production workloads for internal teams.
- Replaced a third-party recommendation vendor with an *in-house k-NN recommender* after benchmarking Gorse against production load, serving from a single non-GPU VM and cutting recommendation spend *60%* (\$2.5K → \$1K/month).

Machine Learning Researcher, Beatoven.ai

Jan 2024 – Jan 2026

Keywords: PyTorch, Diffusion Transformers, LoRA, Distributed Training, MLOps, Data Orchestration

Bangalore, India

Specializing in generative audio models, large-scale data infrastructure, and foundation model training and optimization

- Designed Beatoven.ai's *Composer* architecture, an embedding-guided loop retrieval and arrangement model for controllable music generation from text and audio.
- Built data curation, preprocessing, and augmentation pipelines for *Maestro*, Beatoven's 2.5B-parameter diffusion foundation model trained on multi-million-track licensed datasets.
- Engineered low-level GPU/CPU parallelization, multithreading, and resumable batch processing for a $7.5\times$ *speedup* in training data preprocessing, cutting dataset turnaround from days to hours.
- Built fine-tuning pipelines for SFX, genre, and artist adaptation; tuned hyperparameters for $2\times$ *faster convergence* ($48 \rightarrow 24$ GPU hours) and improved audio quality.
- Standardized distributed training and inference for in-house embedding-extraction models into modular pipelines reused across research teams.

AI Developer (Full-Stack & Data Systems), Beatoven.ai

July 2021 – Dec 2023

Keywords: Flask, React, MongoDB, Python, REST APIs, Microservices, NLP

Bangalore, India

Specializing in full-stack development, large-scale data ingestion, and metadata automation for music AI systems

- Built Beatoven.ai's *artist portal and data ingestion system*: a full-stack Flask + React platform for large-scale artist submissions, metadata management, human annotation workflows, and internal ops dashboards.
- Designed and deployed a lightweight *NLP model* for metadata extraction from unstructured text, cutting manual tagging overhead $> 70\%$ and enabling high-quality dataset curation for early music generation models.
- Built asynchronous ingestion and cleaning pipelines handling *thousands of artist submissions*, integrated directly with Beatoven's composition and embedding systems.

Founder & Lead Developer, Tarab Instruments

June 2024 – Present

Keywords: Embedded Systems, Audio DSP, PCB Design, Analog Circuits

Bangalore, India

Boutique hardware studio building custom MIDI controllers, Eurorack modules, and guitar pedals

- Founded and run a hardware studio taking products end-to-end (analog circuit design, PCB layout, firmware, and design for manufacturability), shipping to musicians and producers.
- Write real-time audio DSP firmware in C++ (DaisySP) for STM32 and Daisy microcontrollers, implementing synthesis and effects under hard low-latency constraints.
- Teach workshops on music electronics, synthesis, and DIY audio hardware.

EDUCATION

Ramaiah University Of Applied Sciences

2020 - 2024

Bachelor of Technology in Computer Science and Engineering

Bangalore, India

Relevant Coursework: Deep Learning, Machine Learning, Natural Language Processing, Computer Vision, Distributed & Parallel Computing, Optimization, Probability & Statistics, Linear Algebra, Digital Signal Processing, System Design

Thesis: *CrateDig*, an offline desktop app (Win/macOS) for searching large audio sample libraries by natural language or reference audio, using embeddings for local similarity search | [gh/CrateDig](#)